

* Ignore fermionic parts of the Action:

$$A = \frac{1}{2\kappa_D^2} \int d^D x \sqrt{g} \left(R - \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \sum_i \frac{1}{2\kappa(n_i)!} e^{a_i \phi} F_{\mu_1 \dots \mu_{n_i}} F^{\mu_1 \dots \mu_{n_i}} \right)$$

$$F_{\mu_1 \dots \mu_{n_i}} = n_i \partial_{[\mu_1} A_{\mu_2 \dots \mu_{n_i}]}$$

$$2\kappa_D^2 = 16\pi G_D$$

E.C.M for g:

$$\delta \sqrt{g} = \frac{1}{2} \frac{1}{\sqrt{g}} \delta g$$

$$\delta(g) = \delta e^{\text{tr}(\ln(g))} = g + \text{tr}(g^{-1} \delta g)$$

$$\delta g = g g^{\mu\nu} \delta g_{\mu\nu}$$

$$\Rightarrow \delta \sqrt{g} = \frac{1}{2} \sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu} = -\frac{1}{2} \sqrt{g} g_{\mu\nu} \delta g^{\mu\nu}$$

$$\delta A_1 = \frac{1}{2\kappa_D^2} \int d^D x g_{\mu\nu} \left(\delta R - \frac{1}{2} \delta g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \sum_i \frac{1}{2\kappa(n_i)!} e^{a_i \phi} F_{\mu_1 \dots \mu_{n_i}} F^{\mu_1 \dots \mu_{n_i}} \right) \delta g^{\mu\nu}$$

$$\delta A_2 = \frac{1}{2\kappa_D^2} \int d^D x \sqrt{g} \left(\delta R - \frac{1}{2} \delta g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \sum_i \frac{1}{2\kappa(n_i-1)!} F_{\mu_1 \dots \mu_{n_i}} F^{\mu_1 \dots \mu_{n_i}} \times \delta g^{\mu\nu} \right)$$

$$\Rightarrow \delta R = \delta(g^{\mu\nu} R_{\mu\nu})$$

$$= \delta g^{\mu\nu} R_{\mu\nu} + g^{\mu\nu} \delta R_{\mu\nu}$$

$$\delta R_{\mu\nu} = \nabla_\mu \Gamma_{\nu\alpha}^\alpha - \nabla_\nu \Gamma_{\mu\alpha}^\alpha$$

which vanishes.

$$\delta A_2 = \frac{1}{2\kappa_D^2} \int d^D x \sqrt{g} \left(R_{\mu\nu} - \frac{1}{2} \partial_\mu \phi \partial_\nu \phi - \sum_i \frac{e^{a_i \phi}}{2(n_i-1)!} F_{\mu_1 \dots \mu_{n_i}} F^{\mu_1 \dots \mu_{n_i}} \right) \delta g^{\mu\nu}$$

$$\therefore -\frac{1}{2} g_{\mu\nu} R + R_{\mu\nu} - \frac{1}{2} \partial_\mu \phi \partial_\nu \phi - \sum_i \frac{e^{a_i \phi}}{2(n_i-1)!} F_{\mu_1 \dots \mu_{n_i}} F^{\mu_1 \dots \mu_{n_i}} = 0$$

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{1}{2} \partial_\mu \phi \partial_\nu \phi + \sum_i \frac{e^{a_i \phi}}{2(n_i-1)!} F_{\mu_1 \dots \mu_{n_i}} F^{\mu_1 \dots \mu_{n_i}}$$

Solve for R:

$$R - \frac{1}{2} D R = \dots$$

$$R(2-D) = \partial_\mu \phi \partial^\mu \phi + \sum_i \frac{e^{a_i \phi}}{(n_i-1)!} F_{\mu_1 \dots \mu_{n_i}} F^{\mu_1 \dots \mu_{n_i}}$$

$$R = \frac{1}{(2-D)} \left(\partial_\mu \phi \partial^\mu \phi + \sum_i \frac{n_i}{(n_i)!} F_{\lambda_1 \dots \lambda_{n_i}} F^{\lambda_1 \dots \lambda_{n_i}} \right)$$

$$\therefore R_{\sigma\lambda} = \frac{1}{2} g_{\sigma\lambda} \frac{1}{(2-D)} \left(\partial_\mu \phi \partial^\mu \phi + \sum_i \frac{n_i}{(n_i)!} F_{\lambda_1 \dots \lambda_{n_i}} F^{\lambda_1 \dots \lambda_{n_i}} \right) \\ = \frac{1}{2} \partial_\sigma \phi \partial_\lambda \phi + \sum_i \frac{e^{a_i \phi} n_i}{2(n_i)!} F_{\mu_1 \dots \mu_{n_i}} F^{\mu_1 \dots \mu_{n_i}}$$

$$R^\mu{}_\nu = \frac{1}{2} \left(\delta^\mu{}_\nu \frac{1}{(2-D)} \partial_\rho \phi \partial^\rho \phi + \partial_\nu \phi \partial^\mu \phi \right) \\ + \sum_i \frac{e^{a_i \phi} n_i}{2(n_i)!} \left(F_{\mu_1 \dots \mu_{n_i}} F^{\mu_1 \dots \mu_{n_i}} \delta^\mu{}_\nu + \frac{\partial_\nu F_{\lambda_1 \dots \lambda_{n_i}} F^{\lambda_1 \dots \lambda_{n_i}}}{(2-D)} \right)$$

* E.O.M of ϕ :

$$\frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} \partial^\mu \phi) - \sum_i \frac{a_i e^{a_i \phi}}{2(n_i)!} F_{\mu_1 \dots \mu_{n_i}} F^{\mu_1 \dots \mu_{n_i}} = 0$$

* E.O.M for $A_{(n_i-1)}$: $F_{(n_i)} = \partial_{[\mu_1} A_{\mu_2 \dots \mu_{n_i}]}$

$$\partial_{[\mu} (\sqrt{-g} e^{a_i \phi} F^{\mu \lambda_2 \dots \lambda_{n_i}}) = 0 \quad (\forall n_i)$$

$$+ \text{ Bianchi } \partial_{[\lambda_1} F_{\lambda_2 \dots \lambda_{n_i+1]} = 0 \\ dF = 0$$

• Note $\rightarrow F_{(1)} = d\phi$

• Work with C.S.-term separately.

$$\text{Recall: } (*\omega)_{\mu_1 \dots \mu_q} = \frac{1}{q!} \epsilon_{\mu_1 \dots \mu_q \nu_1 \dots \nu_{D-q}} \omega^{\nu_1 \dots \nu_{D-q}} \\ = \frac{1}{q!} \sqrt{-g}^{-1} \epsilon \dots$$

$$*F_{(n_i)} = G_{(D-n_i)} \Rightarrow *^2 F_{(n_i)} = *G_{(D-n_i)}$$

$$\text{Define } \hat{F}_{(D-n_i)} \equiv e^{a_i \phi} G_{(D-n_i)}$$

$$* \hat{F}_{(D-n_i)} = (*\hat{F})^{\mu_1 \dots \mu_{n_i}} = \frac{1}{(D-n_i)!} (\sqrt{-g})^{-1} \epsilon^{\mu_1 \dots \mu_{n_i} \nu_1 \dots \nu_{D-n_i}}$$

$$(*\hat{F})^{\mu_1 \dots \mu_{n_i}} = e^{a_i \phi} (*G)^{\mu_1 \dots \mu_{n_i}} = e^{a_i \phi} F^{\mu_1 \dots \mu_{n_i}}$$

$$\dots a^{\alpha\beta} \sqrt{-g} F^{M_1 \dots M_{n_i}} = \frac{1}{(D-n_i)!} \epsilon^{M_1 \dots M_D} \tilde{F}_{M_1 \dots M_{n_i}}$$

which we can replace in E.O.M. to obtain Brane:

$$dF = 0, \quad *G = F$$

$$\Rightarrow \begin{cases} d*G = 0 \\ dF = 0 \end{cases}$$

P-brane Solutions:

(P-brane) action as a generalisation of the N.G. string action:

$$S_p = -T_p \int d\mu_p \quad \left| \begin{array}{l} T_p: \text{P-brane tension} \\ d\mu_p: (p+1)\text{-dim vol. element} \end{array} \right.$$

where:

$$d\mu_p = \sqrt{-\det(G_{\alpha\beta})} d^{p+1}\sigma$$

$$G_{\alpha\beta} = g_{\mu\nu}(x) \partial_\alpha x^\mu \partial_\beta x^\nu, \quad \alpha, \beta = 0, \dots, p$$

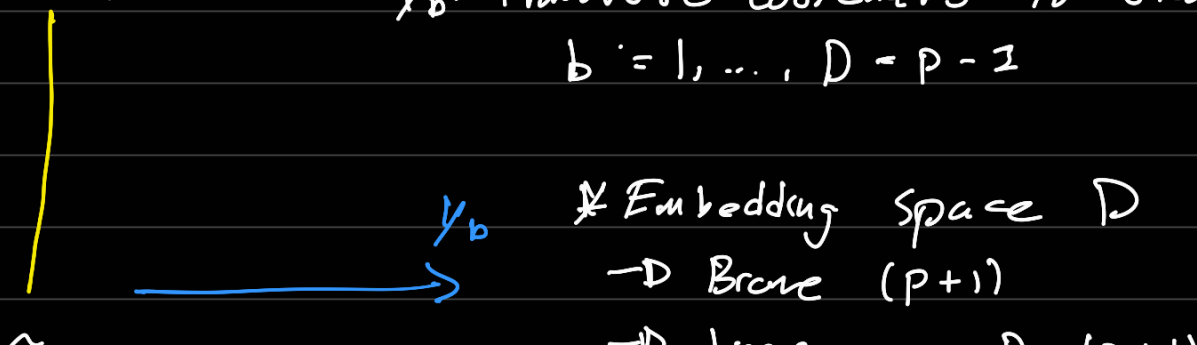
$$[T_p] = \frac{\text{mass}}{(\text{length})^p}$$

• Diffeo invariant.



* Brane world volume (p+1)-dim

Coordinates on B_p : (t, x_j) $j = 1, \dots, p$
 p. (Brane)
 x_b : transverse coordinates to brane
 $b = 1, \dots, D-p-1$



(t, x_i)

- Assume the brane is static, and so we assume all fields are invariant under

$$t \mapsto t + a, \quad a \in \mathbb{R} \quad (\text{E. conservation})$$

$$x_j \mapsto x_j + a_j, \quad a_j \in \mathbb{R} \quad \forall j \quad (\text{man. conservation})$$

Q: How do the fields have non-trivial solutions if there are no dynamics?

— well, I guess the Kerr solution makes sense

* World volume has $SO(p)$ symmetry acting on x_i

→ transverse symmetry: $SO(p-1)$

* line element is time-reversal invariant: $t \mapsto -t$
and so no terms linear in dt

Embedding space metric: Define: $d \equiv D - p - 1$

$$ds^2 = -B^2 dt^2 + C^2 \sum_{j=1}^p (dx_j)^2 + E^2 \sum_{a=1}^d (dy_a)^2 + F^2 \left(\sum_{\alpha} y_{\alpha} dx_{\alpha} \right)^2$$

B, C, E, F : can only depend on $r \equiv \sqrt{\sum_{\alpha} y_{\alpha} y_{\alpha}}$

→ Convert to spherical: $(r, \theta_1, \dots, \theta_{d-1})$ (transverse coords)

$$\begin{cases} y_a = r \sin(\theta_1) \dots \sin(\theta_{a-1}) \cos(\theta_a) & a = 1, \dots, d-1 \\ y_d = r \sin(\theta_1) \dots \sin(\theta_{d-1}) \end{cases}$$

$$\begin{aligned} * dy_{\alpha} &= dr \left(\begin{array}{c} \cos(\theta_1) \dots \cos(\theta_{d-1}) \\ \sin(\theta_1) \dots \sin(\theta_{d-1}) \end{array} \right) + r \sum_{i=1}^{d-1} \cos(\theta_i) \sin(\theta_{i+1}) \dots \sin(\theta_{d-1}) d\theta_i \\ &\quad - r \sin(\theta_1) \dots \sin(\theta_{d-1}) d\theta_{d-1} \\ dy_d &= dr \left(\begin{array}{c} \cos(\theta_1) \dots \cos(\theta_{d-1}) \\ \sin(\theta_1) \dots \sin(\theta_{d-1}) \end{array} \right) \sin(\theta_d) + r \sum_{i=1}^{d-1} \cos(\theta_i) \sin(\theta_{i+1}) \dots \sin(\theta_{d-1}) d\theta_i \end{aligned}$$

$$\sum_{\alpha=1}^d dx_{\alpha} dy_{\alpha} = dr^2 + r^2 d\Omega_{d-1}^2$$

where

$$d\Omega_{d-1}^2 = d\theta_1 + \sin^2(\theta_1) d\theta_2^2 + \dots + \sin^2(\theta_1) \sin^2(\theta_2) \dots \sin^2(\theta_{d-2}) d\theta_{d-1}^2$$

and

$$\sum_{\alpha=1}^d y_{\alpha} dy_{\alpha} = r dr$$

for terms:

$$y_1 dx_1 = (r \cos(\theta_2)) (dr \cos(\theta_1) - d\theta_1 \sin(\theta_1) r) \\ = r^2 c_1^2 dr - r^2 c_1 s_1 d\theta_1$$

$$y_2 dy_2 = (r \sin(\theta_1) \cos(\theta_2)) (dr \sin(\theta_1) \cos(\theta_2) + r \cos(\theta_1) \cos(\theta_2) d\theta_1 \\ - r \cos(\theta_1) \sin(\theta_2) d\theta_2) \\ = r dr \sin^2(\theta_1) \cos^2(\theta_2) + r^2 \sin(\theta_1) \cos(\theta_1) \cos^2(\theta_2) d\theta_1 \\ - r^2 s_1 c_2 c_1 s_2 d\theta_2$$

telescope... use induction.

∴

$$\star ds^2 = -B^2 dt^2 + c^2 \sum_{i=1}^P dx_i^2 + F^2 (dr^2 + r^2 d\Omega_{d-1}^2) + F^2 r^2 dr^2 \star$$

$SO(D-p-1)$ invariance implies ds^2 should be invariant under $r \mapsto r'$

$$y'_{\alpha} = R_{\alpha}^{\beta} y_{\beta}, \quad dy'_{\alpha} = \frac{\partial R_{\alpha}^{\beta}}{\partial y_{\gamma}} y_{\beta} dy_{\gamma} + R_{\alpha}^{\beta} dy_{\beta}$$

$$dy'_{\alpha} = \frac{\partial y_{\gamma}}{\partial y'_{\alpha}} dy_{\gamma} = R_{\alpha}^{-1 \gamma} dy_{\gamma}$$

$$(r')^2 = \sum_{\alpha} y'_{\alpha} y'_{\alpha}$$

$$dr'^2 = \sum_{\alpha} y'_{\alpha} dy'_{\alpha}$$

$$(r')^2 (dr')^2 = \left(\sum_{\alpha} y'_{\alpha} dy'_{\alpha} \right)^2$$

$$F(r') = F(r) + \frac{\partial F}{\partial r} r' + \mathcal{O}(r^2)$$

$$F^2 = F^2(r) + 2F(r) \frac{\partial F}{\partial r} r' + \mathcal{O}(r^2)$$

$$F(r) = F(r) + 2F(r) \frac{1}{dr} r + 10r + O(r^3)$$

→ Can set $F=0$ somehow from this...

$$ds^2 = -B^2 dt^2 + C^2 \sum_{j=1}^p dx_j^2 + E^2 \sum_{a=1}^d (dy_a)^2$$

$$= -B^2 dt^2 + C^2 () + E^2 (dr + r^2 d\Omega_{d-1}^2)$$

* p-branes couple to (p+1) gauge fields:
 → these must also satisfy the above symmetries:

$$(F_t)_{i_1 \dots i_p a} = \epsilon_{i_1 \dots i_p} \partial_a A(r) \text{ or } (A_t)_{i_1 \dots i_p} = (-1)^{p+1} \epsilon_{i_1 \dots i_p} A(r)$$

Since $\epsilon_{i_1 \dots i_p}$ is $SO(p)$ invariant: i.e. 1
 proof: $SO(p) = \{ M \in GL(p) \mid M^T M = \mathbb{1}, \det(M) = 1 \}$

thus:

$$G_{i_1 \dots i_p} \mapsto M_{i_1}^{j_1} \dots M_{i_p}^{j_p} \epsilon_{j_1 \dots j_p}$$

$$= \det(M) \epsilon_{i_1 \dots i_p} = \epsilon_{i_1 \dots i_p} \quad \checkmark$$

$$(F_t)_{i_1 \dots i_p r} = \epsilon_{i_1 \dots i_p} \frac{dA(r)}{dr} \leftarrow \text{Electric Ansatz}$$

* Another possible Ansatz:

$$F^{a_1 \dots a_{p+1}} = \frac{e^{a_1 \phi} \epsilon^{a_1 \dots a_{p+1}} \partial_{a_{p+1}} \tilde{A}(r)}{\sqrt{-g}}$$

(Magnetic Ansatz)

Maxwell: (p=0) brane: particle:

• Couples to a (p+1)=1 form Gauge field A_μ with field strength $F^{(2)}$

$$\begin{cases} (F_0)_a = \partial_a A(r) \\ F_{ab} = \epsilon_{abc} \partial_c \tilde{A}(r) \end{cases}, \phi=0$$

* Assume, for simplicity $a_i = a$, $\phi = \phi(r)$

• Also that only rank (p+2) field strength is non-zero

Step 1 | Compute components of the Ricci tensor of $M^D = B^p \times X^d$

$$\Gamma_{\rho\lambda}^{\sigma} = \frac{1}{2} \tilde{g}^{\sigma\alpha} (\partial_{\rho} \tilde{g}_{\lambda\alpha} + \partial_{\lambda} \tilde{g}_{\rho\alpha} - \partial_{\alpha} \tilde{g}_{\rho\lambda})$$

$$\tilde{g}_{\mu\nu}(r) = \begin{pmatrix} -B^2 & 0 \\ 0 & C^2 \\ & & E^2 \end{pmatrix} \Rightarrow \tilde{g}^{\alpha\beta}(r) = \begin{pmatrix} -B^{-2} & & \\ & C^{-2} & \\ & & E^{-2} \end{pmatrix}$$

$$\Gamma_{t\lambda}^{\sigma} = \frac{1}{2} g^{\sigma\alpha} (\partial_{\lambda} g_{t\alpha} - \partial_{\alpha} g_{t\lambda})$$

$$\begin{aligned} &= \frac{1}{2} g^{\sigma\alpha} (\partial_{\lambda} g_{tt}) \delta_t^{\sigma} - \frac{1}{2} g^{\sigma\alpha} \partial_{\alpha} g_{t\lambda} \\ &= \frac{1}{2} (-B^{-2}) \partial_{\lambda} (-B^2) - \frac{1}{2} g^{\sigma i} \partial_i g_{t\lambda} \\ &= g^{\sigma t} (B^{-2} \partial_{\lambda} B^2) \frac{1}{2} - \frac{1}{2} g^{\sigma i} \partial_i g_{t\lambda} \\ &= \delta_t^{\sigma} \partial_{\lambda} \ln(B) - \frac{1}{2} g^{\sigma i} \partial_i g_{t\lambda} \end{aligned}$$

$$\begin{array}{l|l} \Gamma_{t\lambda}^t = \partial_{\lambda} \ln(B) & \Gamma_{ty_d}^{\sigma} = \partial_{y_d} \ln(B) \\ \Gamma_{t\lambda}^{x_i} = 0 & \Gamma_{tt}^{y_d} = \partial_{y_d} \ln(B) \\ \Gamma_{tx_j}^{\sigma} = 0 & \end{array}$$

$$\begin{aligned} \Gamma_{\sigma\lambda}^t &= -\frac{1}{2} B^{-2} (\partial_{\sigma} g_{\lambda t} + \partial_{\lambda} g_{\sigma t} - \partial_t g_{\sigma\lambda}) \\ &= -\frac{1}{2} B^{-2} () \\ &= 0 \text{ unless } \sigma, \lambda = t/y_d \end{aligned}$$

$$\Gamma_{ty_d}^{y_d} = -\frac{1}{2} g^{y_d y_d} \partial_{y_d} g_{t\lambda} = \partial_{y_d} \ln(E)$$

$$\Gamma_{ty_d}^t = \partial_{y_d} \ln(B) \quad | \quad \Gamma_{ty_d}^{y_d} = \partial_{y_d} \ln(E)$$

$$\begin{aligned} \Gamma_{\sigma\mu}^{x_i} &= \frac{1}{2} g^{x_i x_j} (\partial_{\sigma} g_{x_j\mu} + \partial_{\mu} g_{x_j\sigma}) \\ &= -\frac{1}{2} C^{-2} (\partial_{\sigma} \end{aligned}$$

$$\sigma, \mu \neq t$$

$$\begin{aligned} \Gamma_{x_k\mu}^{x_j} &= -\frac{1}{2} C^{-2} (\partial_{\mu} g_{x_j x_k}) \\ &= \partial_{\mu} \ln(C) \end{aligned}$$

$$\Gamma_{x_i y_d}^{x_j} = \partial_{y_d} \ln(C)$$

$$R_{\mu\nu} = \partial_\sigma \Gamma_{\mu\nu}^\sigma - \partial_\nu \Gamma_{\mu\sigma}^\sigma + \Gamma_{\sigma\tau}^\sigma \Gamma_{\mu\nu}^\tau - \Gamma_{\mu\tau}^\sigma \Gamma_{\sigma\nu}^\tau$$

$\therefore$

$$R^\mu_\nu = g^{\mu\lambda} (\partial_\sigma \Gamma_{\lambda\nu}^\sigma - \partial_\nu \Gamma_{\lambda\sigma}^\sigma + \Gamma_{\sigma\tau}^\sigma \Gamma_{\lambda\nu}^\tau - \Gamma_{\lambda\tau}^\sigma \Gamma_{\sigma\nu}^\tau)$$

$$\begin{aligned} R^t_t &= g^{tt} (\partial_\sigma \Gamma_{t\nu}^\sigma - \partial_\nu \Gamma_{t\sigma}^\sigma + \Gamma_{\sigma\tau}^\sigma \Gamma_{t\nu}^\tau - \Gamma_{t\tau}^\sigma \Gamma_{\sigma\nu}^\tau) \\ &= -B^{-2} \left(\sum_a \partial_{y_a} \partial_{y_a} \ln(B) - \cancel{\partial_t^2} + \dots \right) \end{aligned}$$

$$\begin{aligned} \Gamma_{\sigma\tau}^\sigma \Gamma_{tt}^\tau &= \Gamma_{\sigma t}^\sigma \partial_{y_d} \ln(B) \\ &= \Gamma_{y_a t}^{y_a} \partial_{y_a} \ln(B) = d \sum_a \partial_{y_a} \ln(B) \partial_{y_a} \ln(B) \end{aligned}$$

$$\begin{aligned} -\Gamma_{tt}^\sigma \Gamma_{\sigma t}^\tau &= -\Gamma_{tt}^\sigma \Gamma_{\sigma t}^\tau - \Gamma_{ti}^i \Gamma_{\sigma t}^i - \Gamma_{ta}^a \Gamma_{\sigma t}^a \\ &= -\Gamma_{tt}^{y_a} \Gamma_{y_a t}^{y_a} \\ &\quad - \sum_a \partial_{y_a} \ln(B) \partial_{y_a} \ln(B) \end{aligned}$$

$$-\Gamma_{ti}^\sigma \Gamma_{\sigma t}^i = -\Gamma_{tx_i}^\sigma \Gamma_{\sigma t}^{x_i}$$

$$-\Gamma_{ty_a}^\sigma \Gamma_{\sigma t}^{y_a}$$

* Don't get E^{-2} term, lets try with forms

$$ds^2 = -B^2 dt^2 + c^2 \sum_{i=1}^p dx_i^2 + F^2 \sum_{a=1}^d (dy_a)^2$$

$$\phi(x^\mu) = \phi(r)$$

$$g_{\mu\nu} = e_\mu^A e_\nu^B \eta_{AB} \quad | \quad e^A = dx^\mu e_\mu^A$$

$$\begin{array}{c} \text{brane} \searrow \quad \swarrow \text{trans.} \\ \underline{A} = (\mu, m) \end{array}$$

$$e^i = dx^j e_j^i \quad | \quad e^m = dy^a e_a^m$$

$$e^t = dt e_t^t = B(r) dt$$

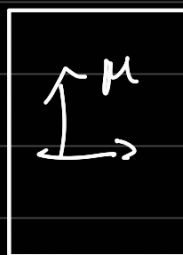
$$e^i = dx^i C(r)$$

$$e^a = dy^a E$$

* Connection 2-form:

$$de^M + \omega^M_N \wedge e^N = 0$$

$$\omega^{ij} = 0 \quad \omega \dots$$



$$\left\{ \begin{aligned} R_t^t &= E^{-2} \left\{ -\sum_a \partial_{y_a} \partial_{y_a} \ln(B) - \sum_a \partial_{y_a} \ln(B) \partial_{y_a} \psi \right\} \\ R_{x_i}^{x_i} &= E^{-2} \delta_i^i \left\{ -\sum_a \partial_{y_a} \partial_{y_a} \ln(C) - \sum_a \partial_{y_a} \ln(C) \partial_{y_a} \psi \right\} \\ R_{y_a}^{y_a} &= \dots \quad (\psi \equiv \ln(B) + p \ln(C) + (d-2) \ln(E)) \end{aligned} \right.$$

* Note if $\psi=0$, then

$$B C^p E^{(d-2)} = 1$$

The above reduces to a simple solⁿ: this preserves

same susy:

* Now: $y_a \rightarrow (r, \theta, \dots)$

$$\partial_a = \frac{\partial r}{\partial y_a} \frac{\partial}{\partial r} \quad \left| \quad \frac{\partial y_a}{\partial r} = \frac{y_a}{r} \right.$$

$$\partial_a B = \frac{y_a}{r} \partial_r B$$

↑ not sure how this is y_a/r and not $\frac{r}{y_a}$

$$\begin{aligned} \partial_b (\partial_a B) &= \partial_a \left(\frac{y_b}{r} B' \right) = \frac{y_b}{r} \frac{\partial}{\partial r} \left(\frac{y_a}{r} B' \right) \\ &= \left(\delta_b^a \frac{1}{r} B' \right) - \frac{y_b}{r} \left(\frac{y_a}{r^2} B' \right) + \frac{y_a y_b}{r^2} B'' \\ &= \frac{y_b}{r} \left(\frac{y_a}{r^2} B' - \frac{y_a}{r^2} B' \right) \end{aligned}$$

$$= \delta_b^a \frac{1}{r} B' + \frac{y_b y_a}{r^2} \left(B'' - \frac{B'}{r} \right)$$

$$\sum_a \partial_a \partial_a B = B'' + \frac{(d-1)}{r} B'$$

$$R_t^t = E^{-2} \left[-\ln(B)'' - \frac{(d-1)}{r} (\ln(B))' - \ln(B)' \psi' \right]$$

$$R_{x_j}^{x_i} = E^{-2} \left[-(\ln(c))'' - \frac{(d-1)}{r} \ln(c)' - \ln(c)' \psi' \right] \delta_j^i$$

etc...

$$R_{y_a}^{y_b} = \delta_a^b R_1 + \frac{y_a y_b}{r^2} R_2$$

In spherical: $R_r^r = \frac{\partial y_a}{\partial r} \frac{\partial r}{\partial y_b} R_{y_a}^{y_b}$

$$= \frac{\partial y_a}{\partial r} \frac{\partial r}{\partial y_b} \left(\delta_b^a R_1 + \frac{y_a y_b}{r^2} R_2 \right)$$

$$\frac{\partial y_a}{\partial r} = \frac{y_a}{r} \quad \left| \quad \frac{\partial r}{\partial y_b} = \frac{r}{y_b} \right.$$

$$R_r^r = \frac{y_a}{y_b} R_b^a = \sum_{a,b} \frac{y_a}{y_b} \left(\delta_b^a R_1 + \frac{y_a y_b}{r^2} R_2 \right)$$

$$= R_1 + \sum_{a,b} \frac{y_a^2}{r} R_2 = R_1 + R_2$$

$$\therefore R_r^r = R_1 + R_2$$

$$R_{\theta_a}^{\theta_b} = R_1 \delta_a^b, \quad \alpha, \beta = 1, \dots, d-1$$

Spherical Coordinates:

$$R_t^t = E^{-2} \left(-\ln(B)'' - \frac{(d-1)}{r} \ln(B)' - \ln(B)' \psi' \right)$$

$$R_{x_j}^{x_i} = E^{-2} \left(-\ln(c)'' - \frac{(d-1)}{r} \ln(c)' - \ln(c)' \psi' \right) \delta_j^i$$

$$R_r^r = E^{-2} \left[-\psi'' + \ln(E)' \psi' - \ln(E)'' - \frac{(d-1)}{r} (\ln(E))' \right. \\ \left. - (\ln(B))'^2 \sim \rho (\ln(c))'^2 - (d-2) (\ln(E))'^2 \right]$$

$$R_a^b = E^{-2} \left(-\ln(E)'' - \frac{(d-1)}{r} (\ln(E))' - \ln(E)' \psi' - \frac{1}{r} \psi' \right) \delta_a^b$$

$$\nabla_\mu \left(\sqrt{-g} e^{a\phi} F^{\mu\lambda_1 \dots \lambda_{d-1}} \right) = 0$$

$$= \partial_\mu \left(\sqrt{-g} e^{a\phi} \epsilon^{\lambda_1 \dots \lambda_{d-1} \mu} \partial^\mu A(r) \right)$$

$$\sqrt{-g} = \sqrt{(r^2)^{d-2} e^{2\rho} E^{2(d-1)}}$$

$$V = \frac{1}{2} (-B^2 - E^2) = -\frac{1}{2} (B^2 + E^2)$$

$$\partial_\mu (B C^p E^d e^{a\phi} \epsilon^{\lambda_1 \dots \lambda_{d-1}} \partial^\mu A(r)) = 0$$

or

$$\partial_\mu (B C^p E^d e^{a\phi} \partial^\mu A(r)) = 0$$

$$\partial_\mu (\dots) + \partial_\alpha (B C^p E^d g^{\alpha\beta} \partial_\beta (A(r))) = 0$$

$$\frac{d}{dr} ((B C^p E)^{-1} e^{a\phi} (E_r)^{d-1} A(r)) = 0$$

=>

$$(B C^p E)^{-1} e^{a\phi} (E_r)^{d-1} A(r)' = Q$$

$$A(r)' = \frac{B C^p E Q}{(E_r)^{d-1}} e^{-a\phi}$$

int. cons of charge

$$\therefore (F_\pm)_{i_1 \dots i_p r} = \epsilon_{i_1 \dots i_p} B C^p E e^{-a\phi} \frac{Q}{(E_r)^{d-1}} *$$

ex: $p=0: (F_\pm)_r = B E e^{-a\phi} \frac{Q}{(E_r)^2}$, $D=4$
(Maxwell)

A: can find after finding B, C, E and ϕ

$$F_{\mu_1 \dots \mu_{p+2}} F^{\mu_1 \dots \mu_{p+2}} = (p+2)! F_{i_1 \dots i_p r} F^{\pm i_1 \dots i_p r}$$

$$= (p+2)! (\epsilon_{i_1 \dots i_p} E^{\pm i_1 \dots i_p}) = \boxed{-(p+2)! e^{-2a\phi} \frac{Q^2}{(E_r)^{2(d-1)}}}$$

Not sure where other term went

$$\epsilon_{\mu_1 \dots \mu_p} = \epsilon^{-1} \epsilon_{i_1 \dots i_p} e^{i_1}_{\mu_1} \dots e^{i_p}_{\mu_p}$$

$$\epsilon^{\mu_1 \dots \mu_p} \epsilon_{\mu_1 \dots \mu_p} = \epsilon_{i_1 \dots i_p} \delta^{i_1}_{j_1} \dots \delta^{i_p}_{j_p} \epsilon^{j_1 \dots j_p}$$

$$= -p!$$

* Other term:

$$F^{\mu_1 \mu_2 \dots \mu_{p+2}} F_{\mu_1 \mu_2 \dots \mu_{p+2}}$$

$$\delta^{i_1}_{j_1} F^{i_1 \mu_2 \dots \mu_{p+2}} F_{i_1 \mu_2 \dots \mu_{p+2}} + F^{a \mu_2 \dots \mu_{p+2}} F_{a \mu_2 \dots \mu_{p+2}} \delta^{i_1}_{j_1} \delta^{i_2}_{j_2}$$

$$+ F^{\pm \mu_2 \dots \mu_{p+2}} F_{\pm \mu_2 \dots \mu_{p+2}} + F^{r \mu_2 \dots \mu_{p+2}} F_{r \mu_2 \dots \mu_{p+2}}$$

$$F^{\mu\lambda_1 \dots \lambda_{p+1}} F_{\nu\lambda_2 \dots \lambda_{p+1}} = (\delta_i^\mu \delta_\nu^i + \delta_t^\mu \delta_\nu^t + \delta_r^\mu \delta_\nu^r) F^{t\lambda_1 \dots \lambda_{p+1}} F_{t\lambda_2 \dots \lambda_{p+1}}$$

$$= (\delta_i^\mu \delta_\nu^i + \delta_t^\mu \delta_\nu^t + \delta_r^\mu \delta_\nu^r) (p+1)! F_{t\lambda_2 \dots \lambda_{p+1}} F^{t\lambda_2 \dots \lambda_{p+1}}$$

$$= -(\delta_t^\mu \delta_\nu^t + \delta_i^\mu \delta_\nu^i + \delta_r^\mu \delta_\nu^r) (p+1)! e^{-2\alpha\phi} \frac{Q^2}{(Er)^{2(d-1)}}$$

all other components vanish

$$R^\mu_\nu = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi$$

$$= \sum_i \frac{1}{2n_i!} e^{\alpha\phi} n_i (\delta_t^\mu \delta_\nu^t + \delta_i^\mu \delta_\nu^i + \delta_r^\mu \delta_\nu^r) (p+1)! \frac{Q^2}{(Er)^{2(d-1)}}$$

$$\Rightarrow \sum_i \frac{e^{\alpha\phi}}{2n_i!} \binom{n_i-1}{D-2} \left((p+2)! e^{-2\alpha\phi} \frac{Q^2}{(Er)^{2(d-1)}} \right)$$

$$R^t_t = - \sum_i \frac{e^{-\alpha\phi}}{2n_i!} \frac{Q^2}{(Er)^{2(d-1)}} \frac{1}{2} \left(\frac{n_i-1}{(D-2)} - n_i (n_i-1)! \right)$$

$$p+2 = n_i \Rightarrow p = n_i - 2$$

$$= - \frac{e^{-\alpha\phi} Q^2}{4(Er)^{2(d-1)}} \sum_{i=1}^d \left(\frac{n_i-1}{(D-2)} - 1 \right)$$

$$d = D - n_t - 1$$

$$n_i = D - d - 1$$

$$\Rightarrow \frac{D-d-2}{(D-2)} = - \frac{1}{(D-2)}$$

Not quite:

$$R^t_t = E^{-2} \left(-(\ln(B))'' - \ln(B)' \left(4^e + \frac{d-1}{r} \right) \right)$$

$$= - \frac{d-2}{2(D-2)} e^{-\alpha\phi} \frac{Q^2}{(Er)^{2(d-1)}}$$

* and likewise for all other terms

oh... we said only $(p+2)$ term was relevant i.e. $n_i = p+2$ and no sum

$$\text{on } \ln(B) \text{ term } \left(\frac{e^{\alpha\phi}}{2n_i!} \right) (p+2) F^{\mu\lambda_2 \dots \lambda_{p+2}} F_{\mu\lambda_2 \dots \lambda_{p+2}}$$

$$= \frac{(p-1)}{(D-2)} F_{\lambda_1 \dots \lambda_{p+2}} F_{\lambda_1 \dots \lambda_{p+2}}$$

Second term:

$$\frac{e^{a\phi}}{2(p+2)!} \left((p+2)(-\delta)(p+1)! e^{-2a\phi} \frac{Q^2}{(Er)^{2(d-1)}} - \frac{(p-1)(p+2)!}{(D-2)} e^{Q^2} \right)$$

$$= \frac{e^{-a\phi} Q^2}{2} \left((-\delta) - \frac{p-1}{D-2} \right)$$

$$R_t^t = - \frac{e^{-a\phi} Q^2}{2} \left(\frac{D-2+p-1}{D-2} \right) = - \frac{e^{-a\phi} Q^2}{2 (Er)^{2(d-1)}} \frac{(d-2)}{(D-2)}$$

as claimed.

* Dilation E.O.M: $\frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} g^{\mu\nu} \phi) - \frac{Q}{2(p+2)!} F_{\lambda_1 \dots \lambda_{p+1}} F_{\lambda_1 \dots \lambda_{p+2}} = 0$

$$\sqrt{g} = (B C^p E) \cdot (Er)^{d-1}$$

$$\frac{1}{\sqrt{g}} \partial_r (\sqrt{g} E^{-2} \phi') = \left(\frac{1}{B C^p E \cdot (Er)^{d-1}} \right) \left(B C^p E (Er)^{d-1} \frac{1}{E^2} \phi' \right)'$$

$$= \frac{1}{E^2} \left[\phi'' + \frac{d-1}{r} \phi' + \phi' \left(\ln(B)' + p \ln(C)' - \ln(E)' + (d-1) \ln(E)' \right) \right]$$

$$= \frac{1}{E^2} \left[\phi'' + \frac{(d-1)}{r} \phi' + \phi' \psi' \right]$$

$$= E^{-2} \left[\phi'' + \phi' \left(\frac{d-1}{r} + \psi' \right) \right]$$

$$\therefore E^{-2} \left(\phi'' + \phi' \left(\frac{d-1}{r} + \psi' \right) \right) = \frac{-a e^{-2a\phi}}{2(p+2)!} \frac{Q^2}{(Er)^{2(d-1)}} (p+2)!$$

$$\therefore E^{-2} \left(\phi'' + \phi' \left(\frac{d-1}{r} + \psi' \right) \right) = - \frac{a}{2} e^{-2a\phi} \frac{Q^2}{(Er)^{2(d-1)}}$$

ϕ E.O.M

* Solve for $\psi = 0$: $B C^p E^{d-2} = 1$, then all the

above reduce to:

$$(i) \frac{d}{dr} (r^{d-1} \ln(B)') = \frac{d-2}{2(D-2)} e^{-a\phi} E^{-2(d-2)} \frac{Q^2}{r^{d-1}}$$

$$(ii) \frac{d}{dr} (r^{d-1} \ln(C)') = \frac{d-2}{2(D-2)} e^{-a\phi} E^{-2(d-2)} \frac{Q^2}{r^{d-1}}$$

$$\begin{aligned}
 \text{(iii)} \quad & -\frac{1}{r^{d-1}} \frac{d}{dr} (r^{d-1} \ln(E)') - (d-2) (\ln(E)')^2 - (p+1) (\ln(C)')^2 \\
 & = \frac{1}{2} \phi'^2 - \frac{d-2}{2(d-2)} e^{-\alpha\phi} E^{-2(d-2)} Q^2 / r^{d-1} \\
 \text{(iv)} \quad & \frac{d}{dr} (r^{d-1} \ln(E)') = -\frac{p+1}{2(d-2)} E^{-2(d-2)} e^{-\alpha\phi} \frac{Q^2}{r^{d-1}} \\
 \text{(v)} \quad & \frac{d}{dr} (r^{d-1} \phi') = -\frac{\alpha}{2} E^{-2(d-2)} e^{-\alpha\phi} Q^2 / r^{d-1}
 \end{aligned}$$

* Impose the brane has $SO(1,p)$ which enlarges the $SO(p)$ symmetry:

$$\begin{aligned}
 \text{Brane: } & ISO^+(d,1) \otimes SO(p) \\
 & \longrightarrow ISO^+(1,p) \otimes SO(d)
 \end{aligned}$$

$$ds^2 = B^2 \left(-dt^2 + \sum_{j=1}^p dx_j^2 \right) + E^2 \sum_{a=1}^d (dy_a)^2$$

$$\therefore B = C$$

as could have been inferred from the first two equations above: in reality we'd get $B \propto C$ and we could redefine t^* s.t. we absorb the α term $B = E^{2-d/p+1}$

$$\psi = 0 \iff B C^p E^{d-2} \rightarrow B^{p+1} E^{d-2} = 1$$

• From this (iii) and (iv) are equal

• Simplify (iii):

$$\begin{aligned}
 & -\frac{1}{r^{d-1}} \frac{d}{dr} (r^{d-1} \ln(E)') - (d-2) (\ln(E)')^2 - (p+1) \left(\frac{2-d}{p+1} \ln(E)' \right)^2 \\
 & = -\frac{1}{r^{d-1}} \frac{d}{dr} (r^{d-1} \ln(E)') - \left((d-2) + \frac{(d-2)^2}{(p+1)} \right) (\ln(E)')^2 \\
 & = \left(\frac{1}{r^{d-1}} \right) \frac{p+1}{2(d-2)} E^{-2(d-2)} e^{-\alpha\phi} \frac{Q^2}{r^{d-1}} - \left((d-2) + \frac{(d-2)^2}{(p+1)} \right) (\ln(E)')^2 \\
 & = \frac{1}{2} \phi'^2 - \frac{d-2}{2(d-2)} e^{-\alpha\phi} E^{-2(d-2)} Q^2 / r^{2(d-1)}
 \end{aligned}$$

$$\frac{1}{2(p-2)} E^{-2(d-2)} e^{-\alpha\phi} \frac{Q^2}{r^{2(d-1)}} \underbrace{[p+1 + d-2]}_{p-2} = ()$$

$$\left(\frac{p+1 + d-2}{p-2} \right) (d-2) (\ln(E)')^2 = \frac{1}{2} \phi'^2 -$$

$$- \frac{(D-2)(D-2)}{(p+1)} (h_1(E)')^2 = \frac{1}{2} \phi'^2 - \frac{e^{-\alpha\phi}}{2} E^{-2(D-2)} \frac{Q^2}{r^{2(D-2)}} \quad (iii)$$

* Since (i), (ii) and (iv) are identical and (iii) above

we have (v) left

$$\frac{d}{dr}(r^{d-1}\phi') = -\frac{\alpha}{2} E^{-2(D-2)} e^{-\alpha\phi} \frac{Q^2}{r^{d-1}}$$

$ISO(1,p) \otimes SO(d)$

$$\frac{d}{dr}(r^{d-1} h_1(E)') = -\frac{(p+1)}{2(D-2)} E^{-2(D-2)} \frac{Q^2}{r^{d-1}}$$

Next: Solve for a theory with no dilation

∴

$$\sqrt{\frac{2(D-2)(D-2)}{(p+1)}} \frac{d}{dr} h_1(E) = E^{-(D-2)} \frac{Q}{r^{d-1}}$$

$$\int E^{(d-3)} dE = \int \frac{Q}{r^{d-1}} dr ()^{-1/2}$$

$$\frac{1}{(d-2)} E^{d-2} = -\frac{Q}{d} \frac{1}{r^{d-2}} () + (d-2)^{-1}$$

$$\therefore E = \left(1 + \sqrt{\frac{(p+1)}{2(D-2)(D-2)}} \frac{|Q|}{r^{d/2}} \right)^{2/(d-2)}$$

$$B = \left(1 + \sqrt{\frac{(p+1)}{2(D-2)(D-2)}} \frac{|Q|}{r^{d/2}} \right)^{-1/(p+1)}$$

* This solution has used the fact that as

$$r \rightarrow \infty \quad E = 1 \quad \text{i.e.}$$

$$X^d \rightarrow M^d$$

* Theory with a dilation: $\frac{d}{dr}(r^{d-1}\phi') = -\frac{\alpha}{2} E^{-2(D-2)} e^{-\alpha\phi} \frac{Q^2}{r^{d-1}}$

and $\frac{d}{dr}(r^{d-1} h_1(E)') = -\frac{(p+1)}{2(D-2)} E^{-2(D-2)} e^{-\alpha\phi} \frac{Q^2}{r^{d-1}}$

$$\text{or } \frac{d}{dr}(r^{d-1}\phi') = \frac{\alpha}{2} \left(\frac{2(D-2)}{p+1} \right) \frac{d}{dr}(r^{d-1} h_1(E)')$$

$$\text{and } \phi(r) = \frac{(p+1)}{2(D-2)} \phi$$

and so. $\lambda(\frac{1}{2}) = -\frac{(p+1)}{a(D-2)} \phi$ | $E = e$

$$-\left(\frac{(D-2)(d-2)}{p+1}\right) \left(\frac{p+1}{a(D-2)}\right)^2 \phi'^2 = \frac{1}{2} \phi'^2 \rightarrow \frac{1}{2} e^{-a\phi} e^{\frac{p+1}{a(D-2)} \phi (-2)(d-2)} \frac{Q^2}{r^{2(d-1)}}$$

$$e^{\left(\frac{-a^2(D-2) - 2(p+1)(d-2)}{(D-2)} \phi\right)}$$

Define,

$$\Delta \equiv \frac{1}{2} a^2 (D-2) + (p+1)(d-2)$$

$$e^{\frac{-2\Delta\phi}{a^2(D-2)}}$$

$$\phi'^2 \left(\frac{(p+1)(d-2) + \frac{1}{2} a^2 (D-2)}{a^2 (D-2)} \right) = \frac{\Delta}{a^2 (D-2)} \phi'^2$$

$$\frac{\Delta}{a^2 (D-2)} \phi'^2 = \frac{1}{2} e^{\frac{-2\Delta\phi}{a^2(D-2)}} \frac{Q^2}{r^{2(d-1)}}$$

$$\sqrt{\frac{2\Delta}{a^2(D-2)}} \phi' = e^{\frac{-\Delta\phi}{a^2(D-2)}} \frac{|Q|}{r^{d-1}}$$

$$e^{\frac{\Delta\phi}{a^2(D-2)}} \phi' = \sqrt{\frac{a^2(D-2)}{2\Delta}} \frac{|Q|}{r^{d-1}}$$

$$\frac{a(D-2)}{\Delta} \frac{d}{dr} \left(e^{\frac{\Delta\phi}{a^2(D-2)}} \right) = a \sqrt{\frac{(D-2)}{2\Delta}} \frac{|Q|}{r^{d-1}}$$

$$\frac{d}{dr} \left(e^{\frac{\Delta\phi}{a^2(D-2)}} \right) = \sqrt{\frac{\Delta}{2(D-2)}} \frac{|Q|}{r^{d-1}}$$

$$e^{\frac{\Delta\phi}{a^2(D-2)}} = 1 + \sqrt{\frac{\Delta}{2(D-2)}} \frac{|Q|}{(d-2) r^{d-2}}$$

$$\phi(r) \xrightarrow{r \rightarrow \infty} 0$$

$$e^{\frac{\Delta\phi}{a^2(D-2)}} = 1 + \frac{1}{(d-2)} \sqrt{\frac{\Delta}{2(D-2)}} \frac{|Q|}{r^{d-2}}$$

$$\Delta = (p+1)(d-2) + \frac{1}{2} a^2 (D-2)$$

$$B = C$$

→ Assuming $ISO^+(1, p) \otimes SO(d)$ symmetry,

$$\bullet ds^2 = B^2 (-dt^2 + \sum_i dx_i^2) + E^2 (dr^2 + r^2 d\Omega_{d-1}^2)$$

$$\bullet B^{p+1} E^{d-2} = 1 \quad | \quad \bullet \psi = 0$$

$$\bullet \ln(E) = \frac{p+1}{a(D-2)} \phi$$

D=5 case:

$$N \equiv 1 + \frac{1}{(d-2)} \sqrt{\frac{\Delta}{2(d-2)}} \frac{|Q|}{r^{d-2}}$$

$$B = N^{-(d-2)/\Delta}$$

$$E = N^{(p+1)/\Delta}$$

$$e^\phi = N^{a(d-2)/\Delta}$$

$$F_{t \dots i p r} = (\pm) \sqrt{\frac{2(d-2)}{\Delta}} \frac{d}{dr} (N^{-1})$$

$$(\pm) = Q/|Q|$$

$$ds^2 = N^{-2(d-2)/\Delta} (-dt^2 + dx_1^2 + \dots + dx_p^2) + N^{2(p+1)/\Delta} (dr^2 + r^2 d\Omega_{d-1}^2)$$

*** (p-brane Solution) ***

$\psi = 0$ (susy preserving)

* If a theory contains a rank $(p+1)$ gauge field $A^{(p+1)}$ hence a $(p+2)$ Field strength $F^{(p+2)}$ the theory also possesses a spherically symmetric p-brane solution with $\psi=0$, $ISO(1,p) \otimes SO(d)$ symmetry.

(p-brane)



Brane Solutions in 11D Supergravity:

$$(\psi_a, e_{\mu}^a, A^{(3)}) \rightarrow F^{(4)}$$

$$\int * F^{(4)} \wedge F^{(4)} \quad \text{and} \quad \int F^{(4)} \wedge F^{(4)} \wedge A^{(3)}$$

C.S.

* C.S. adds to the action with

$$\epsilon^{\mu_1 \dots \mu_4} F_{\mu_1 \mu_2} F_{\mu_3 \mu_4}$$

For $F_{t i s i s a} = \epsilon_{i s i s} \partial_a A(r)$, the above is zero since $\epsilon^{t \dots t \dots} = 0$ so we have the full supergravity solution.

$$\psi \wedge F^{(4)} = 1 d(3) \quad | \quad \psi \wedge F^{(4)} \wedge A^{(3)} = 1 d(6)$$

$$\left. \begin{array}{l} A_{(3)} \rightarrow M_2 - \text{brane} \\ B_{(6)} \rightarrow M_5 - \text{brane} \end{array} \right\} \underline{\underline{11D - \text{SUGRA}}}$$

M2-brane: $D=11$, $p=2$, $d=8$, $a=0$

• Note this theory is simpler since it contains no Dilaton

$$\Delta = (p+1)(d-2) + \frac{1}{2}a^2(p-2) = 18$$

$$\Delta = 18$$

$$r^2 = y_1^2 + \dots + y_8^2$$

$$N_2 = 1 + \frac{101}{6r^5}$$

$$dy_1^2 + \dots + dy_8^2$$

↓

$$ds^2 = N_2^{-2/3} (-dt^2 + dx_1^2 + dx_2^2) + N_2^{1/3} (dr^2 + r^2 d\Omega_7^2)$$

• 4-form field strength

$$(F_4)_{12a} = \pm \frac{\partial}{\partial y_a} (N_2^{-1}), \quad i = a, \dots, 8$$

M5-brane: look at the Dual Solution

$$D=11, \quad \tilde{D}=5, \quad \tilde{p}=5, \quad \tilde{a}=0, \quad \tilde{\Delta}=18$$

$$N_5 = 1 + \frac{101}{3r^3}, \quad k \equiv \frac{101}{3} \quad | \quad r^2 = y_1^2 + \dots + y_5^2$$

$$F_{1\dots 5a} = \frac{\partial}{\partial y_a} (N_5^{-1})$$

$$F_{a_1\dots a_4b} = \pm \epsilon_{a_1\dots a_4b} \frac{\partial}{\partial y_b} (N_5^{-1}), \quad a_1, \dots, a_4, b = 1, \dots, 5$$

$$ds^2 = N_5^{-1/3} (-dt^2 + dx_1^2 + \dots + dx_5^2) + N_5^{2/3} (dy_1^2 + \dots + dy_5^2)$$

M-theory pp-wave solⁿ: (M-wave)

$$ds^2 = -(1-k)dt^2 + (1+k)dx_1^2 - 2kdt dx_1 + ((dx_2)^2 + \dots + dx_{p-1}^2)$$

$$1+k = 1 + k/r^{p-4} \quad | \quad r^2 = (x_2)^2 + \dots + (x_{p-1})^2$$

- Purely a gravitational solution, hence the 3 and 6 terms vanish.

Kaluza-Klein Monopole:

$$ds^2 = -dt^2 + (dx_1)^2 + \dots + (dx_{p-1})^2 + H^{-1} \left(dz + \sum_{i=1}^3 A_i dx_i \right)^2 + H \sum_{i=1}^3 (dx_i)^2$$

$$F_{ab} = 2 \partial_{[a} A_{b]} = \epsilon_{abc} \partial_c H \quad \left| \quad H = 1 + \frac{Q}{r} \right.$$

$$\partial_a \partial_b H = 0, \quad z = x^{10}$$

- Line elements in the (z, y_a) directions are $\mathbb{R} \times$ Taub-NUT spaces for which the z -direction must be compactified

$$S^1: \quad L = \text{period} \quad \tau = 4\pi Q$$

$$* D=11 \text{ on } AdS_4 \otimes S^7$$

$$\text{Isometry group: } SO(2,3) \otimes SO(8)$$

$$SO(2,3)/\text{conf} \cong ISO(4,2)$$

$$AdS_{\tilde{n}} \quad \Lambda = -3d, \quad \eta^m, \quad m = 1, \dots, \tilde{n}+1$$

$$\eta^m \eta_{mn} \eta^n = -\frac{1}{d^2} \quad | \quad \eta_{nm} = (-1, 1, \dots, 1, -1)$$

$$ds^2_{\tilde{n}} = -(1 + d^2 r^2) dt^2 + (1 + d^2 r^2)^{-1} dr^2 + r^2 d\Omega_{\tilde{n}-2}^2$$

- $AdS_{\tilde{n}}$: pos. curvature so, like a sphere

$$S^n: \quad SO(n+1) \text{ symmetry group:}$$

$$AdS_{p+2} \doteq SO(2, p+1) \quad p+2 = \tilde{n}$$

- $AdS_{p+2} \times S^{D-p-2} : SO(2, p+1) \times SO(p-1)$
- Dual field / brane has

$$\Lambda_{D-p-2} \times S^{p+2} : SO(2, D-p-1) \times SO(p+3)$$

$$+ \text{has } AdS_{2+2} \otimes S^{11-2-2} = AdS_4 \otimes S^7$$

$$\begin{cases} AdS_4 \otimes S^7 : SO(2, 3) \otimes SO(8) \\ AdS_7 \otimes S^4 : SO(2, 8) \otimes SO(5) \quad (\text{D=11}) \end{cases}$$

$\hookrightarrow \hat{F}_{\mu_1 \dots \mu_7}^{(\tau)} \simeq C \in_{\mu_1 \dots \mu_7}$

(Freund-Rubin Sol^{ns})

- Preserve all susy.

* Intersecting brane solutions:

→ Intersecting M2-branes:

- Brane volumes:

$$(M_2)_1 : (X_0, X_1, X_2) \quad 0 \ 1 \ 2 \ \dots$$

$$(M_2)_2 : (\tilde{X}_0, \tilde{X}_3, \tilde{X}_4) \quad 0 \ \dots \ 3 \ 4 \ \dots$$

$$ds^2 = (H_1 H_2)^{1/3} \left[-(H_1 H_2)^{-1} dt^2 + H_1^{-1} (dx_1^2 + dx_2^2) + H_2^{-1} (dx_3^2 + dx_4^2) + (dy_1^2 + \dots + dy_6^2) \right]$$

$$F_{t12a} = \frac{c_1}{2} \partial_a (H_1^{-1}) \quad \Bigg| \quad F_{t34a} = \frac{c_2}{4} \partial_a (H_2^{-1})$$

$$a=1, \dots, 6, \quad c_i = \pm 1, \quad i=1, 2$$

- Harmonic functions are altered $\Bigg| \quad H_i = 1 + \frac{c_i}{r_i^4}$

$$r_i^2 = \sum_a (y_a - y_a^i)^2$$

y_a^i : positions of the 2-brane in the transverse space.

- Preserves 8 of 32 susy

M2 brane configuration: 0 1 2 . . .
0 . . 3 4 . . .

(2)

(1)

* M2-brane Intersecting M5-brane:

$$ds^2 = H_1^{2/3} H_2^{1/3} \left[H_1^{-1} H_2^{-1} (-dt^2 + dx_1^2) + H_1^{-1} (dx_2^2 + dx_3^2 + dx_4^2 + dx_5^2) + H_2^{-1} (dx_6^2) + (dy_1^2 + dy_2^2 + dy_3^2 + dy_4^2) \right]$$

* Note: For every coordinate in the brane direction we have a factor of H_i in front of it: $H_i(dx_i^2 \dots)$
 When the branes overlap, the coordinates will contain both (or more) $H_i H_j (dx^2 \dots)$

0	1	2	3	4	5	·	...	$F_{6abc} = \frac{c_1}{2} \epsilon_{abc} \partial_c H_1$ $F_{716a} = \frac{c_2}{a} \frac{\partial_c H_2}{H_2^2}$
0	1	·	·	·	·	6	...	

$c_i = \pm 1, i=1,2$

* Intersecting M5-branes:

$$ds^2 = (H_1 H_2)^{2/3} \left[(H_1 H_2)^{-1} (-dt^2 + dx_1^2 + dx_2^2 + dx_3^2) + H_1^{-1} (dx_4^2 + dx_5^2) + H_2^{-1} (dx_6^2 + dx_7^2) + (dy_1^2 + dy_2^2 + dy_3^2) \right]$$

$$F_{67ab} = \frac{c_1}{2} \epsilon_{abc} \partial_c H_1$$

$$F_{81ab} = \frac{c_2}{2} \epsilon_{abc} \partial_c H_2$$

$$0 \ 1 \ 2 \ 3 \ 4 \ 5 \ \cdot \ \dots$$

$$0 \ 1 \ 2 \ 3 \ \cdot \ 6 \ 7 \ \dots$$

* General story Here: In $D=11$ the single solⁿ for each brane has a term

$$ds_p^2 = \dots + H^{-\alpha} (dy_1^2 + \dots + dy_{D-p-1}^2)$$

factoring out this $H^{-\alpha}$ we find a conformal factor:

$$ds^2 = H \left(H^{-1} (-dt^2 + dx_1^2 + \dots + dx_{p+1}^2) + (dx_1^2 + \dots) \right)$$

this same form holds exactly as above for intersecting branes, we simply must know what α is for each brane.

Of course

$$\alpha = \frac{2(p+1)}{D-2} \quad \Delta = (p-1)(d-2) + \frac{1}{2}a^2(d-2)$$

$$a=0 \text{ for } D=11$$

* For all of the above we do not take the fermionic fields into account and so, classically

$$\delta\psi = 0, \text{ Hence}$$

$$0 = D_\mu(\hat{\omega})\epsilon + \frac{1}{12^2} (\gamma_{\mu}^{\nu_1 \dots \nu_4} - 8 \delta_{\mu}^{\nu_1} \gamma^{\nu_2 \nu_3 \nu_4}) \hat{F}_{\nu_1 \nu_2 \nu_3 \nu_4} \epsilon$$

$\Rightarrow$ # of spinor solutions is the # of susy parameters preserved.

Brane Solutions in 10D Maximal Supergravity:

$$\text{IIA: } ([1000]_8 \oplus [0010]_8) ([1000]_8 \oplus [0001]_8)$$

$$\left. \begin{array}{l} A_{\mu}^{(1)} \rightarrow F^{(2)} = dA^{(1)} \\ C_{\mu\nu}^{(3)} \rightarrow F^{(4)} = dC^{(3)} \end{array} \right\} \begin{array}{l} \mathbb{R} \otimes \mathbb{R} \\ \text{Sector} \end{array}$$

$$\left(\begin{array}{l} \text{NS} \otimes \text{NS} \\ \text{Sector} \end{array} \right) A^{(2)} \rightarrow F^{(3)} = dA^{(2)}$$

	F	$*F$
$\mathbb{R} \otimes \mathbb{R}$	$F^{(2)}$ $F^{(4)}$	$G^{(8)}$ $G^{(6)}$
$\text{NS} \otimes \text{NS}$	$F^{(3)}$	$G^{(7)}$

$$2-D \quad 1 \quad 0 \quad 1 \quad D-D$$

$$\bullet q_{p+2} = M \left(\frac{3-p}{2} \right) \quad \Bigg| \quad M = \begin{cases} 2 & NS-NS \\ -1 & NS-NS \end{cases}$$

$$p = 1, 2, 3, \quad \sigma = 4$$

• C.S. terms vanish for all solutions which we look at.

p-brane: $F^{(p+2)}$

$p = 0, 1, 2, 4, 5, 6$ brane solutions in $11D$

Note, for all p above $\Delta = 16$

$$\begin{aligned} \Delta &= (p+1)(d-2) + \frac{1}{2} a^2 (p-2) \\ &= (p+1)(d-2) + \frac{1}{2} \left(\frac{3-p}{2} \right)^2 8 \\ &= (p+1)(D-p-3) + (3-p)^2 \\ &= (p+1)(7-p) + \\ &= -p^2 + 6p + 7 + 9 - 6p + p^2 \\ &= 16 \end{aligned}$$

$$\Delta = 16$$

p-brane solution:

$$ds^2 = N^{-2(7-p)/16} () + \frac{2(p+1)}{16} ()$$

$$\begin{aligned} ds^2 &= N^{-\frac{(p+1)}{8}} \left[N^{\frac{(p+1)}{8} + \frac{p-7}{8}} () + () \right] \\ &= N^{-\frac{(p+1)}{8}} \left[N^{\frac{2p-6}{8}} () + () \right] \end{aligned}$$

$$ds^2 = N_p^{-\frac{(p+1)}{8}} \left[N_p^{\frac{p-3}{8}} (-dt^2 + dx_1^2 + \dots + dx_p^2) + (dy_1^2 + \dots + dy_{6-p}^2) \right]$$

$$\ast N_p = 1 + \frac{|Q|}{(7-p)r^{7-p}}, \quad e^\phi = (N_p)^{a_{p+2}/2}, \quad A_{1\dots p+1} = \pm (N_p^{-1} - 1) \quad \ast$$

↳ "optimal"

NS5NS: $p = 1, \quad p = 5$ branes

$\hookrightarrow A^{(2)} \quad A^{(6)}$

$p=1$ brane is a string solution: F1-brane

$p=5$: NS5-brane

R0R: $p = 0, 2, 4, 6$

$$\underbrace{\Delta^{(1)} \quad A^{(3)} \quad A^{(5)} \quad A^{(7)}}_{}$$

D_p-branes

→ Note we have D branes for all but a D₂ brane

$$\underline{p=8}: A^{(9)} \Rightarrow F^{(10)} = dA^{(9)} \sim E^{(10)}$$

$$\text{or } F^{(10)}_{\mu_1 \dots \mu_{10}} = C E_{\mu_1 \dots \mu_{10}}$$

• We can consistently add this to the theory as it is a cosmological constant term Λ_{10}

$$\rightarrow \int d^{10}x \, e F_{\mu_1 \dots \mu_{10}} F^{\mu_1 \dots \mu_{10}}$$

which has E.O.M:

$$\partial_\mu \left(\frac{\delta \mathcal{L}}{\delta \partial_\mu A \dots} \right) + \frac{\delta \mathcal{L}}{\delta A \dots} = 0$$

$$\partial_\mu (E^{\mu_1 \dots \mu_{10}} F_{\mu_1 \dots \mu_{10}}) = 0$$

$$E^{\mu_1 \dots \mu_{10}} F_{\mu_1 \dots \mu_{10}} = C$$

$$\therefore F_{\mu_1 \dots \mu_{10}} = E_{\mu_1 \dots \mu_{10}} C$$

Note the $E^{\mu_1 \dots \mu_{10}}$ comes from the fact

$$\text{that } \frac{\delta F_{\mu_1 \dots \mu_{10}}}{\delta \partial_{\alpha_1} A_{\mu_2 \dots \mu_{10}}} = \frac{\delta [\mu_1 \dots \mu_{10}]}{\delta \alpha_1 \delta \alpha_{10}}$$

$$= -\frac{1}{10! i} \epsilon_{\alpha_1 \dots \alpha_{10}} E^{\mu_1 \dots \mu_{10}}$$

Thus, on-shell:

p=8-brane yields a C.C. term.

$$\boxed{\int d^{10}x \, e \Lambda_{10}}$$

Q? Why doesn't the C.C. term emerge (both here and in GR) naturally? Is our inherent physical assumptions causing $\Lambda=0$ before we do our analysis?

Aside: Adding a C.C. Λ to the string Action leads to inconsistent Classical E.O.M.

$$S_\sigma = -\frac{T}{2} \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X_\mu + \Lambda \int d^2\sigma \sqrt{-h}$$

$\rightarrow$ E.O.M:

$$\frac{2}{\sqrt{-h}} \frac{\delta S}{\delta h^{\alpha\beta}} = -T \left(\partial_\alpha X^\mu \partial_\beta X_\mu - \frac{1}{2} h_{\alpha\beta} \left(h^{\gamma\delta} \partial_\gamma X^\mu \partial_\delta X_\mu \right) \right) - \Lambda h_{\alpha\beta} = 0$$

$$\Lambda h_{\gamma\delta} h^{\gamma\delta} = T \left(\frac{1}{2} h_{\gamma\delta} h^{\gamma\delta} - 1 \right) h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X_\mu$$

$$2\Lambda = 0 \Rightarrow \Lambda = 0$$

* However, the sigma model form of a p-brane P=1 requires a cosmological constant term.

Proof:

$$S_\sigma = -\frac{T_p}{2} \int d^p\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X_\mu + \Lambda_p \int d^p\sigma \sqrt{-h}$$

wait... this doesn't give a global Λ term...
 this is Λ on the brane, not Λ in the spacetime.
 ... (see Becker book $\nearrow$)

$\rightarrow$ D8 brane did not occur from analysis due to the fact that it is not dynamical

D8-brane solⁿ which preserves susy:

$$ds^2 = N_8^{9/8} \left(N_8^{-1} (-dt^2 + dx_1^2 + \dots + dx_8^2) + (dx_9^2) \right)$$

$$A_{1\dots 9} = (N_8)^{-1} - 1, \quad e^4 = (N_8)^{-3/4}$$

Dp brane solutions: BPS solⁿ' of IIA Super

Type IIB: $N = (2, 0)$

$$F_{\pi\pi} = (\{1000\}_3 \oplus \{0010\}_3)^2$$

NS 2-form

$$\underline{NS \otimes NS}: \begin{matrix} [2000]_8 & , & [0000]_8 & & [0100]_8 \\ e_n^\alpha & & \phi & & A^{(2)} \\ \oplus (Duk) & & & & \hat{A}^{(6)} \end{matrix}$$

$$\underline{R \otimes R} \quad \begin{matrix} [0100]_8' & [0000]_8' & [0020] \\ \hat{A}^{(2)} & \chi & A^{(4)} \\ \oplus DUKs: & \hat{A}^{(6)} & \hat{A}^{(8)} \end{matrix} \quad \begin{matrix} \text{I} \\ A^{(8)} \end{matrix} \quad \begin{matrix} \text{(self-dual)} \\ \hat{A}^{(4)} \end{matrix}$$

• In the $R \otimes R$ sector χ acts as the potential for a gauge field, i.e. $F^{(1)} = d\chi$

$$\underline{NS \otimes NS - branes}: p = 1, 5$$

$$\underline{R \otimes R - branes}: p = -1, 1, 3, 5, 7$$

$$\rightarrow \text{Dilaton coupling: } q_{p+2} = \eta \frac{(3-p)}{2} \quad \eta = \begin{cases} 1 & R \otimes R \\ -1 & NS \otimes NS \end{cases}$$

$$\Delta = 16$$

$$N_p = 1 + |Q|/(7-p)r^{7-p}$$

$$C^p = N_p^{-\eta(p-3)/4}$$

$$A_{1\dots p-1} = \pm (N_p)^{-1} - 1$$

$$ds_p^2 = N_p^{(p+1/2)} \left[N_p^{-1} (-dt^2 + dx_1^2 + \dots + dx_p^2) + (dy_1^2 + \dots + dy_d^2) \right]$$

* West claims that $p=7$ brane exists in the $NS \otimes NS$ sector of IIB however I don't think this is correct.

$p=5$

NS $\otimes$ NS: $p=1$ string coupled to a $A^{(2)}$ gauge field which is the B_{mn} field we usually find in S.E:

$$S_B = \frac{2}{4\pi\alpha'} \int d\sigma E^{\alpha\beta} \partial_\alpha x^\mu B_{mn} \partial_\beta x^\nu$$

R $\otimes$ R: $p = -1, 1, 3, 5, 7$ branes

• $p=1$: Here this is a D_1 -string since it contains $R \otimes R$ B.C.s

$\begin{pmatrix} F_1 \\ D_2 \end{pmatrix}$: transform as a doublet ($\mathbb{Z}^2$) of $SU(1,1) \cong SL(2, \mathbb{R})$

- Easily seen from the fact that rank 2-gauge fields transform as such under the above group.

Instanton: $p = -1$, which lives at a point in spacetime.

* Just as in the type IIA case, the R & R sector has branes of odd p for all but $\underline{p=9}$,

$\underline{p=9}$: Space-filling brane

$\rightarrow$ Go back to Sol^n : $E=0$, and we find

$$ds^2_4 = N (-dt^2 + dx_1^2 + \dots + dx_9^2) \quad | \quad N: \text{constant}$$

* Type IIB also has the pure gravitational solutions

(i) pp-waves

(ii) KK-monopoles

* All NS-brane, D_p -brane & grav Sol^n preserve half of the susy.

$\rightarrow$ Also have $AdS_5 \otimes S^{D-5} Sol^n$:

- Take self-dual field strength given by

$$F_{\mu_1 \dots \mu_5} = d\epsilon_{\mu_1 \dots \mu_5} \quad | \quad F_{a_1 \dots a_5} = d\epsilon_{a_1 \dots a_5}$$

and all other fields besides e_μ^a also vanish.

$$R_{\mu\nu}^m = -\frac{c^2}{2} \delta_{\mu\nu}^m, \quad R_b^a = \frac{c^2}{2} \delta_b^a$$

$$\begin{aligned} & \frac{1}{2(S^4)} \left(5 d^2 (-4!) \delta_{\mu\nu}^m - \frac{9}{8} \delta_{\mu\nu}^m (-5! d^2) \right) \\ &= \frac{1}{2} \delta_{\mu\nu}^m d^2 \left(\frac{1}{2} - 1 \right) = -\frac{1}{4} d^2 \delta_{\mu\nu}^m \end{aligned}$$

→ Solution: $AdS^5 \otimes S^5$ with isometry group

$$SO(2,4) \otimes SO(6)$$

(Preserves all 32 supercharges)

Brane Charges and the preservation of susy:

* p-brane sol have energy densities: (M, Q)
mass, charge

• Compute mass + charge via surface integrals

$$Q_e = \frac{1}{2\kappa_D^2} \int_{S^{D-p-2}} *F \quad \left(\begin{array}{l} \text{electric charge} \\ \text{of } D_p\text{-brane} \end{array} \right)$$

• Integral is taken over a sphere at infinity that is in an asymptotic region of space at large radius r .

→ i.e. over the Boundary ∂M^{D-p-1}

$$*F \rightarrow (*F)_c dM \quad \left| \begin{array}{l} dM = r^{D-p-2} d\theta_1 \wedge (sin\theta_1 d\theta_2 \wedge \dots \wedge d\theta_{D-p-2}) \\ (*F)_c = (*F)_{\theta_1 \dots \theta_{D-p-2}} \end{array} \right.$$

$$(*F)_{\theta_1 \dots \theta_{D-p-2}} = \frac{1}{p!} \epsilon_{\theta_1 \dots \theta_{D-p-2} t_1 \dots t_p} F^{t_1 \dots t_p r}$$

$$\text{But we know } F^{t_1 \dots t_p r} = \epsilon^{t_1 \dots t_p r} Q / r^{D-p-2}$$

$$= - \epsilon_{\theta_1 \dots \theta_{D-p-2}} Q / r^{D-p-2}$$

r^{D-p-2} cancels that in the measure

$$\text{thus, } Q_e = \frac{Q}{2\kappa_D^2} \Omega_{D-p-2} \quad \left(\begin{array}{l} D_p\text{-brane elec.} \\ \text{charge} \end{array} \right)$$

$$Q_m = \frac{1}{2\kappa_D^3} \int_{S^{D+2}} F$$

$\left(\begin{array}{l} D_p\text{-brane Mag-} \\ \text{charge} \end{array} \right)$

• Can replace $S^{p+2} \rightarrow \partial M^{p+3}$

* Consider the space the branes live on as T^p , so that the brane is wrapped once around T .

11D: M2 - Electrical charge
M5 - Magnetic charge

• For the M2 brane: 3D, thus we need a S^2 sphere to enclose it $S^2 \cong \underbrace{T^2 \times \dots \times T^2}_7$

* All the same analysis for charged Black holes

* E.O.M of Gauge field:

$$d * F_{(n)} + F_{(n)} \wedge F_{(n)} = 0$$

$\Rightarrow$ linear terms such as $F_{(n)} \wedge F_{(n)}$ do not contribute on the Boundary

* Magnetic charge $dF = 0$ (Bianchi) and arise from nontrivial topology.

* ex: point particle in 4D

$$p^\mu = (E, 0, 0, 0)$$

$$Q_e = \frac{1}{2\kappa_D^2} \int_{S^2} *F = \frac{1}{2\kappa_D^2} \int r^2 d\Omega \underline{E} \cdot \hat{r}$$

$$*F = (*F)_{\theta\phi} d\theta \wedge (sin\theta d\phi)$$

$$E_r = F_{\theta r}, \quad \underline{E} = (e/r^2) \hat{r}$$

$$\text{and so } Q_e = e$$

$$\Rightarrow \int d\Omega r^{-2} = 2\kappa_D^2 \quad \text{normalized as } \underline{S^2}.$$

M2 brane: $Q_e = \frac{1}{2\kappa_D^2} \int_{S^7} *F_{(n)}$

$$*F_{(n)} = \frac{1}{\sqrt{-g}} \epsilon_{\mu_1 \dots \mu_6 \mu_7 \mu_8 \mu_9 \mu_{10}} F_{\mu_7 \mu_8 \mu_9 \mu_{10}} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_6}$$

$$F_{(n)} = dA_{(3)}, \quad S^7: \text{Asymptotic space at } \infty$$

$$(t, x_1, x_2) \text{ on } M_2$$

$$(x_3, \dots, x_{10}) \text{ on } S^7$$

$$F_{(n) t i i_2 a} = \epsilon_{i_1 i_2} \partial_a A(r)$$

$$\text{or } Q_e^{M_2} = \frac{1}{2\kappa_D^2} \int_{S^7} r^7 d\Omega_7 \frac{|Q|}{(r)^7}$$

$$Q_e^{M_2} = |Q| \left(\frac{1}{2\kappa_D^2} \int_{S^7} d\Omega_7 \right) = |Q|$$

* For the M5-brane: (static & planar)

$$(t, x^1, \dots, x^5)$$

Bounded by a 4-sphere S^4

$$Q_m^{M_5} = \frac{1}{2\kappa_D^2} \int_{S^4} F$$

• All of the above assumes zero sources.

$$\partial_m (\sqrt{-g} e^{a_i \phi} F^{a_1 a_2 \dots a_n}) = 0$$

Sources

* P-branes move according to an action that contains

(i) brane-world volume term

(ii) Coupling to a rank $(p+1)$ gauge field of the background supergravity theory.

$$\hookrightarrow \sim \int d^{p+1} \xi A_{M_1 \dots M_{p+1}} j^{M_1 \dots M_{p+1}}$$

$j^{M_1 \dots M_{p+1}}$: Corresponding current (topological)

$$j^{M_1 \dots M_{p+1}} = g_p dx^{M_1} \wedge \dots \wedge dx^{M_{p+1}}$$

$X^m(\xi)$: embedding coordinates of the p-brane into M_{amb}



q_d : charge

$$S_{\text{brane}} = S_{\text{w.v.}} + S_{\text{gauge}}$$

$$\bullet S_{\text{w.v.}} = T_p \int d^{p+1} \xi (\dots)$$

T_p : brane tension

$$S_{\text{gauge}} = g \int d^{p+1} \xi \epsilon \dots$$

* A p-brane has two coupling constants (T_p, g) which are generally independent.

→ From susp.: $T_p = g$

As such, the E.O.M of A will contain the top current times a δ -fn

$$\bar{J}^{\mu_1 \dots \mu_{p+1}} = \int d^{p+1} \xi \delta(x^\mu - X^\mu(\xi)) j^{\mu_1 \dots \mu_{p+1}}$$

* Sources means: E.O.M

$$\star d \star F = \bar{J}$$

or

$$d \star F = \star \bar{J}$$

$$\bullet \text{ Thus, } Q = \frac{2}{2\kappa_D^2} \int_{\partial M_{D-p-1}} \star F = \frac{1}{2\kappa_D} \int_{M_{D-p-1}} d \star F = \frac{1}{2\kappa_D} \int_{M_{D-p-1}} \star \bar{J}$$

$$\text{or } Q = \frac{1}{2\kappa_D^2} \int_{M_{D-p-1}} \star \bar{J}$$

Supersymmetry:

* To see if a solⁿ is susy invariant just vary via susy $\delta \epsilon$:

$$\left. \begin{array}{l} \psi_m = 0 \\ \chi^\alpha = 0 \end{array} \right\} \text{ must vanish for classical solutions}$$

and hence $\delta\psi_m = \delta\lambda^a = 0$ and so,
 $\delta(\text{Bosons}) = 0$ since they are all prop-
to fermions.

• ex: D=11 sugra: $\delta\psi_m = 0$

$$0 = \frac{1}{\kappa} D_m(\tilde{\omega})\epsilon + \frac{1}{12^2} \left(\int_m^{\nu_1 \dots \nu_4} - 8 \delta_m^{\nu_1} \int^{\nu_2 \nu_3 \nu_4} \right) \hat{F}_{\nu_1 \dots \nu_4} \epsilon$$

$$\frac{1}{\kappa} \left(\partial_m - \frac{1}{4} \omega_m^{mn} \gamma_{mn} \right) \epsilon + \frac{1}{12^2} \left(\Gamma - 8 \Gamma^S \right) 4 \partial_{[m} A_{n_2 m_2 n_2]} \epsilon = 0$$

must solve for a given brane solⁿ

ex: (M2-brane) $F_{t12a} = \pm \partial_a(N_2^{-1})$

$$\frac{1}{\kappa} \left(\partial_m - \frac{1}{4} \omega_m^{mn} \gamma_{mn} \right) \epsilon + \frac{1}{12^2} \left(\int_m^{t12a} - 8 \delta_m^t \int^{12a} \right) \pm \partial_a N_2 \epsilon$$

$$\begin{aligned} \partial_a N_2^{-1} &= -1 \frac{1}{N_2^2} \frac{|Q|}{6} \frac{\partial}{\partial y^a} (r^{-6}) \\ &= \frac{|Q|}{N_2^2 r^7} \frac{\partial}{\partial y^a} (r) \end{aligned}$$

$$\frac{\partial r}{\partial y^a} = \frac{1}{2} \frac{1}{\sqrt{y^2 \dots}} \frac{\partial}{\partial y^a} (y^2 \dots) = \frac{y_a}{r}$$

$$\partial_a(N_2^{-1}) = \frac{|Q|}{N_2^2 r^8} y_a$$

$$\frac{1}{\kappa} \left(\partial_m - \frac{1}{4} \omega_m^{mn} \gamma_{mn} \right) \epsilon + \frac{1}{12^2} \left(\int_m^{t12a} - 8 \delta_m^t \int^{12a} \right) \frac{|Q| y_a}{N_2^2 r^8} \epsilon$$

$$[g_{mn}] = \begin{pmatrix} -N_2^{-2/3} & & & \\ & N_2^{-2/3} & & \\ & & 2 & \\ & & & N_2^{1/3} \\ & & 0 & & 8 \end{pmatrix}$$

$$e_m^a \eta_{ab} e_n^b$$

$$1 - N_2^{-1/3} \quad 0 \quad \backslash$$

$$[e_m^m] = \begin{pmatrix} N_2^{-1/3} \\ \frac{1}{2} \\ 0 \end{pmatrix} \begin{pmatrix} N_2^{-1/6} \\ 8 \end{pmatrix}$$

$$\omega_{\mu}^{mn} = \omega_{\mu} e_a^{\mu} e_n^{\mu}$$

Plug in values ... come back to this and do calculation:

M2-brane: $E = N_2^{-1/6} E_0$ s.t. $\Gamma^{012} E = \pm E$

at $r \rightarrow \infty$

$$E(r) = E_0 \quad \text{since} \quad N_2 \rightarrow 1$$

* Oh... for $\mu = r$

$$\frac{1}{h} [\partial_r - \frac{1}{2} \omega_r^{mn} \gamma_{mn}] E + \frac{1}{12^2} (\Gamma_r^{1234}) \frac{1}{N_2^{2/3}} E = 0$$

so this is easier to solve:

$$\omega_{\mu ab} = \frac{1}{2} (-C_{\mu ab} + C_{ab\mu} - C_{b\mu a})$$

$$C_{\mu\nu}^a = 2 \partial_{[\mu} e_{\nu]}^a$$

M5-brane: $E = N_5^{-1/12} E_0$ | $\Gamma^{012345} E = \pm E$

• E : spinor in $\text{Spin}(1, D-1)$

$$\longrightarrow \text{Spin}(1, p) \otimes \text{Spin}(D-p-1)$$

$$E = \eta_2 \otimes \eta_2$$

→ Thus, basically the same procedure as in the compactified cases.

* Can also look at the susy algo. Since

$$\delta(\text{fields}) = 0 \quad \text{and} \quad \delta \sim \epsilon^\alpha Q_\alpha, \text{ then}$$

$$\text{also} \quad \{Q_\alpha, Q_\beta\} \bar{\epsilon}_\beta^B = 0$$

• Thus, we can look at zero modes of $\{Q, Q\}$

(i) $Z = 0$ ($\forall$ central charges)

$$\{Q_\alpha, Q_\beta\} = (\Gamma^m C^{-1})_{\alpha\beta} P_m$$

then

$$\{Q_\alpha, Q_\beta\} \psi = (\Gamma^m C^{-1})_{\alpha\beta} P_m \psi^\beta$$

or

$$(\Gamma^m C^{-1})_{\alpha\beta} P_m \psi^\beta = (\Gamma^m)_\alpha{}^\gamma (C^{-1})_{\gamma\beta} P_m \psi^\beta$$

$$= (\Gamma^m)_\alpha{}^\gamma (P_m \psi)_\gamma$$

$$= (\Gamma^m P_m)_\alpha{}^\gamma \psi_\gamma$$

$$P_\alpha{}^\gamma \psi_\gamma = 0$$

and so $\det(P) = 0 = (p^2)^6$ or

$p^2 = 0$ (i.e. massless)

$\rightarrow$ Boost to the frame $P^m = m(1, \pm 1, 0, \dots)$

$$\{Q_\alpha, Q_\beta\} = m(\Gamma^0 \pm \Gamma^1) C^{-1})_{\alpha\beta}$$

$$(\Gamma^0)^2 = 41$$

$$\begin{cases} \gamma^0 \gamma^1 + \gamma^1 \gamma^0 = 0 \Rightarrow \gamma^0 \gamma^1 = -\gamma^1 \gamma^0 \\ \Gamma^{01} = \frac{1}{2} (\gamma^0 \gamma^1 - \gamma^1 \gamma^0) = \gamma^0 \gamma^1 \end{cases}$$

$$\boxed{\{Q_\alpha, Q_\beta\} = m(\Gamma^0 (1 \pm \Gamma^{01}) C^{-1})_{\alpha\beta}}$$

thus, we must have $\{Q_\alpha, Q_\beta\} \epsilon_p = 0$, or

$$\boxed{(1 \pm \Gamma^{01}) \epsilon_p = 0}$$

* Since $(\Gamma^{01})^2 = 1$, then the above is a projection operator: half susy preserved for zero central

charge

$\rightarrow$ pp-wave solutions

$$ds^2 = -(1-K)dt^2 + (1+K)dx_1^2 + 2K dt dx_1 + (dx_2^2 + \dots + dx_{D-1}^2)$$

$$\text{where } 1+K = 1 + \frac{r^2}{r_0^2 - 4}, \quad r^2 = x_2^2 + \dots + x_{D-1}^2$$

$\rightarrow$ 3, 6 form gauge fields vanish

1. D = 11

* Now include the M2-brane with central extension z^{mn}

$$\text{Rank}(z^{mn}) = 1 \Rightarrow z^{12} \neq 0$$

$$\{Q_\alpha, Q_\beta\} = (m \Gamma^0 (1 + b \Gamma^{012}) C^{-1})_{\alpha\beta}$$

* Since: $p^\mu = (m, 0, 0, \dots)$ (massive)

$$\begin{aligned} \{Q_\alpha, Q_\beta\} &= (\gamma^\mu C^{-1})_{\alpha\beta} p_\mu + (\Gamma^{mn} C^{-1})_{\alpha\beta} z_{mn} \\ &= m (\Gamma^0 C^{-1})_{\alpha\beta} + (\Gamma^{12} C^{-1})_{\alpha\beta} z_{12} \end{aligned}$$

$$\left\{ \begin{aligned} \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu &= 2 \eta^{\mu\nu} \mathbb{1}_{32} \\ \gamma^\mu \gamma_\mu + \gamma_\mu \gamma^\mu &= 2 \delta^\mu_\mu \\ \gamma^0 \gamma_0 + \gamma_0 \gamma^0 &= 2 \end{aligned} \right.$$

$$\{Q_\alpha, Q_\beta\} = m \Gamma^0 (1 + \Gamma^{012} m^{-1} z_{12}) C^{-1})_{\alpha\beta}$$

let $b \equiv z_{12} m^{-1}$ and Γ^0

$$\{Q_\alpha, Q_\beta\} = m (\Gamma^0 (1 + b \Gamma^{012}) C^{-1})_{\alpha\beta}$$

(M2-brane only)

again, we must have $\text{tr}[\ln(1 + b \Gamma^{012})]$

$$\det(1 + b \Gamma^{012}) = e$$

$$\ln(1+x) = b \Gamma^{012} - \frac{b^2 \mathbb{1}}{2} + \frac{b^3 \Gamma^{012}}{3} + \dots$$

but

$$\text{tr}(\Gamma^{012}) = 0$$

$$\text{tr}(\mathbb{1}) = ?$$

$$\dim([\mathfrak{so}(10)]_6) = 2^5 = 32$$

Since reps $\mathfrak{so}(10)$

$$\text{tr}(\mathbb{1}) = 32$$

thus

$$\text{tr}[\ln(1 + b \Gamma^{012})] = 16 \ln(1 - b^2)$$

$$16 \ln(1 - b^2)$$

$$16 \ln(1 - b^2)$$

$$\therefore \det(1 + b \Gamma^{012}) = e^{b^2} = (1 - b^2)^{-10} = 0$$

thus, to preserve susy

$$b = \pm 1 \Rightarrow (1 \pm \Gamma^{012}) \epsilon_p = 0$$

or $\Gamma^{12} = \pm m$

preserves half susy

$$\underline{32} \rightarrow \underline{16} \text{ susy}$$

* For p-branes, the source current $J^{M_1 \dots M_{p+1}}$ becomes the central charges in susy and the preserved susy satisfies:

$$\int^{0 \dots p} G = \pm \epsilon$$

$$\uparrow$$

$$\delta \psi_\mu = 0$$

• Also applies for type IIA & IIB

M5-brane: let's again assume only one component of $Z^{(5)}$ is non-zero (as above). Not sure why, guess we'll see in the next section.

$$Z^{12345} \equiv Z$$

→ Massive spin reps of $SO(10)$: $[00001]_{10}$
 $P^\mu = (m, 0, \dots, 0)$

$$\bullet \{Q_\alpha, Q_\beta\} = (\Gamma^0 C^{-1})_{\alpha\beta} m + Z_{12345} (\Gamma^{12345} C^{-1})_{\alpha\beta}$$

$$= m (\Gamma^0 (1 + m^{-1} Z \Gamma^{012345}) C^{-1})_{\alpha\beta}$$

→ Basically, as above the Γ^i 's from $D=11$ break into the highest γ on the brane which acts as a projection.

→ Again $Z = \pm m$?

* Susy alg. allows us to determine/derive Bounds involving the mass & central charges of the

branes

$$\{Q_\alpha, Q_\beta\} = \dots$$

multiply by two grassmann even spinors η
i.e.

$$\begin{aligned} \eta &\in \mathbb{C} \otimes \text{Spin}(1,10) \\ \bar{\eta}^\alpha \{Q_\alpha, Q_\beta\} \eta^\beta &= \bar{\eta}^\alpha (Q_\alpha Q_\beta + Q_\beta Q_\alpha) \eta^\beta \\ &= (\bar{\eta} Q) (\eta Q) + (\eta Q) (\bar{\eta} Q) \\ &= 2(\bar{\eta} Q) \end{aligned}$$

where $\bar{\eta} = \eta^\dagger \Gamma^0$, thus

$$\begin{aligned} (\bar{\eta} Q)^\dagger &= (\eta^\dagger \Gamma^0 Q)^\dagger = Q^\dagger (\Gamma^0)^\dagger \eta = -Q^\dagger \Gamma^0 \eta \\ &= -\bar{Q} \eta = \bar{\eta} Q \end{aligned}$$

$$\Rightarrow \underline{(\bar{\eta} Q) \in \mathbb{R}} \Rightarrow \underline{(\bar{\eta} Q)^2 \geq 0}$$

$$2(\bar{\eta} Q)^2 = \bar{\eta} (\Gamma^M P_M + \dots + \dots) \eta \geq 0$$

ex: $D=11$, M2-brane

$$\underbrace{\bar{\eta}^\alpha}_{\neq 0} \left[(\Gamma^M C^{-1})_{\alpha\beta} P_M + Z^{ab} (\Gamma_{ab} C^{-1})_{\alpha\beta} \right] \underbrace{\eta^\beta}_{\neq 0} \geq 0$$

$$P_M = (m, 0, \dots, 0), \quad Z^{01} \neq 0$$

$$m (\Gamma^0 C^{-1})_{\alpha\beta} + Z^{12} (\Gamma^{12} C^{-1})_{\alpha\beta} \geq 0$$

mult by C

$$m \Gamma^0 + Z^{12} \Gamma^{12} \geq 0$$

$$\left\{ \begin{array}{l} \text{eigenvalues of } \nearrow \text{ must be of the form} \\ M \pm |Z^{12}| \\ \text{ } \end{array} \right. \quad \text{ } \quad \Gamma^{012} \epsilon = \pm \epsilon$$

or

$$\begin{aligned} (m \mathbb{1} + Z^{12} \Gamma^{012}) \epsilon &= (m \pm |Z^{12}|) \epsilon \\ &= (m \pm |Z^{12}|) \epsilon \end{aligned}$$

* M2 brane solution saturates this bound
meaning $m = \pm |Z^{12}|$

BPS-branes : Belong to short supermultiplets
or BPS representations